GOVERNMENT OF KHYBER PAKHTUNKHWA

KHYBER PAKHTUNKHWA TEXTBOOK BOARD PESHAWAR

OFFICIAL PRESCRIBED E-CURRICULUM TEXTBOOK

CHEMISTRY CLASS 10

CLASS / GRADE: 10th (Matric)

Subject: Chemistry | Language: English

AUTHENTIC CURRICULUM SYLLABUS & CHAPTER BLUEPRINTS

· Total Complete Curriculum Chapters: 8 Units

· Designed as per National Curriculum & KPK BISEP Standards

· Contains complete theory, conceptual exercises, MCQs & solutions

· Fully synchronized with ExamForge AI Examination Paper Generator

Published by: Khyber Pakhtunkhwa Textbook Board, Phase-V, Hayatabad, Peshawar

Approved by: Directorate of Curriculum and Teacher Education (DCTE) Abbottabad, KP

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UNIT 1: CHEMICAL EQUILIBRIUM: DYNAMIC EQUILIBRIUM, LAW OF

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Chemical Equilibrium: Dynamic Equilibrium, Law of Mass Action, Eq

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

· Direct Exam Paper Integration:

This chapter is directly connected to ExamForge AI's Blueprint Builder.
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UNIT 2: ACIDS, BASES AND SALTS: ARRHENIUS, BRONSTED-LOWRY,

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Acids, Bases and Salts: Arrhenius, Bronsted-Lowry, Lewis Concepts

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 3: ORGANIC CHEMISTRY: CLASSIFICATION, FUNCTIONAL GROU

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Organic Chemistry: Classification, Functional Groups, Homologous

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 4: HYDROCARBONS: PREPARATION AND CHEMICAL REACTIONS O

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Hydrocarbons: Preparation and Chemical Reactions of Alkanes, Alk

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 5: BIOCHEMISTRY: CARBOHYDRATES, PROTEINS, LIPIDS, NUC

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Biochemistry: Carbohydrates, Proteins, Lipids, Nucleic Acids (DNA &

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 6: THE ATMOSPHERE: LAYERS (TROPOSPHERE, STRATOSPHERE)

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of The Atmosphere: Layers (Troposphere, Stratosphere), Greenhouse

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 7: WATER: HARDNESS OF WATER, WATER-BORNE DISEASES, WA

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Water: Hardness of Water, Water-borne Diseases, Water Purificatio

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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UNIT 8: CHEMICAL INDUSTRIES: METALLURGY OF COPPER, SOLVAY'

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1. Learning Outcomes & Key Competencies:

· Understand core fundamental concepts and principles of Chemical Industries: Metallurgy of Copper, Solvay's Process, Urea I

· Develop analytical comprehension, factual recall, and problem solving skills.

· Relate theoretical knowledge to everyday practical applications in Pakistan.

2. Theoretical Foundation & Textbook Synopsis:

This unit covers the authentic curriculum prescribed by the KPK Directorate of Curriculum.
Students are guided through structured topic explanations, standard definitions, solved examples,
and foundational laws verified by subject specialists of BISE Peshawar.

3. Official Textbook Exercise Questions (BISEP Pattern):

[Section A - Objective]: Choose the correct option from the given 4 choices.

[Section B - Short Questions]: Answer 8 conceptual questions with scientific reasoning.

[Section C - Detailed Analysis]: Answer 3 comprehensive theoretical and numerical questions.

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